

Lending Club Analysis

Overview:

- Lending Club is a peer-to-peer lending platform that connects borrowers with investors, enabling individuals to obtain loans without traditional financial institutions. It provides an online marketplace where borrowers can apply for loans and investors can fund them based on risk and return preferences.

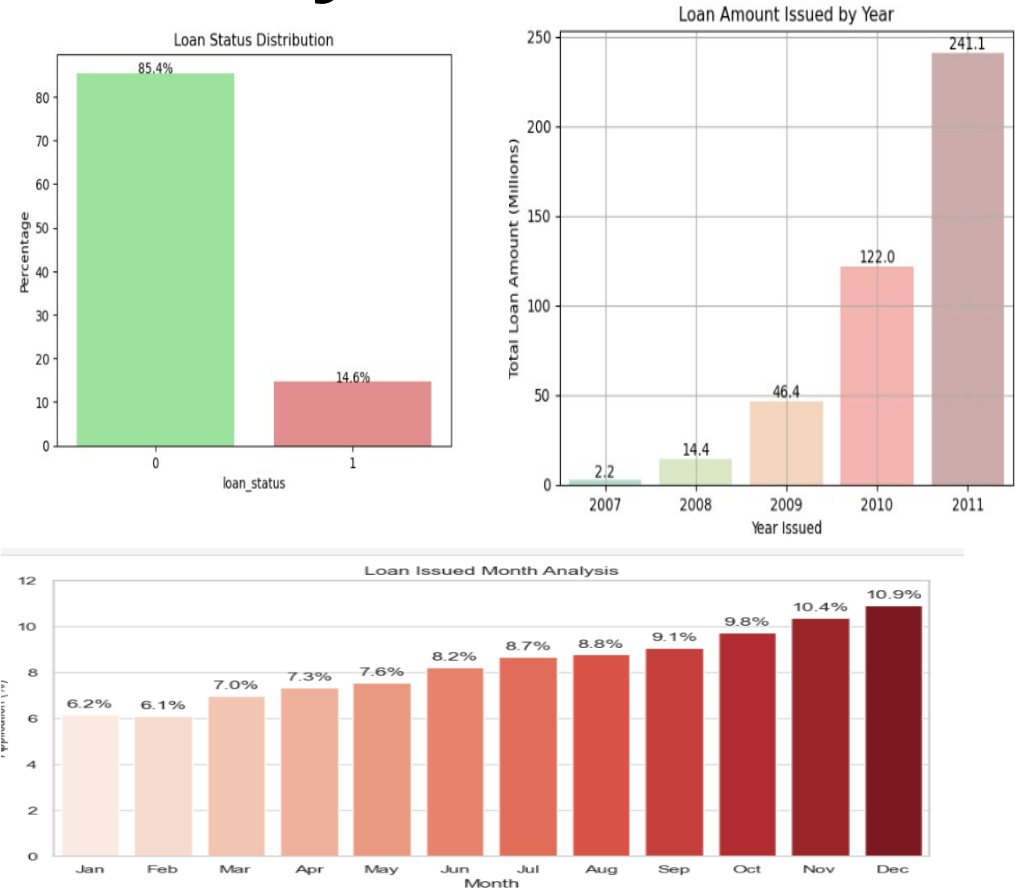
Problem Statement:

- Lending Club faces difficulty in accurately assessing the credit risk of borrowers, especially since the loans are unsecured.
- There is a high risk of loan default, which can lead to financial losses for investors.
- It is essential to identify reliable borrowers by analyzing applicant-related factors to minimize default risk.

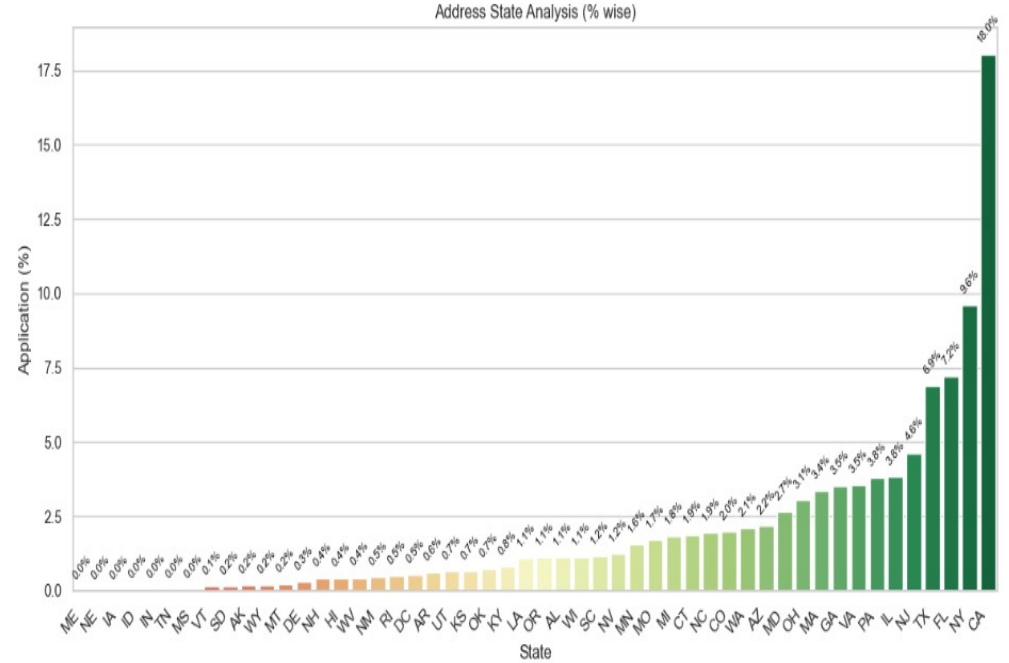
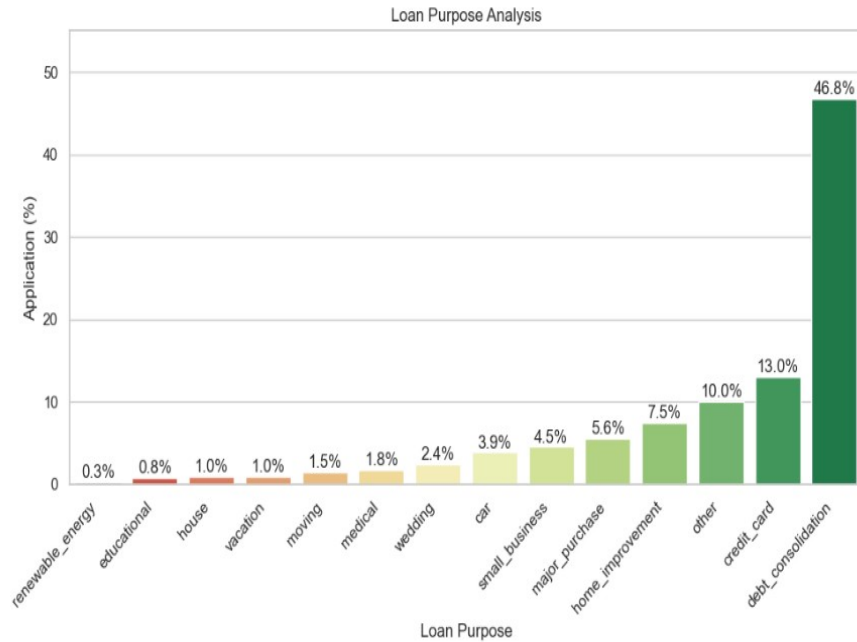
Univariate Analysis

Univariate Analysis: Insights

- Overall Default Rate stands at 14.59%.
- Approval rate of loans jumped by around 78% from the year 2010 to 2011
- Approval Rate of loans is higher during the Holiday Season (November & December).
- Approved Loan amount almost doubled between years' 2010 and 2011



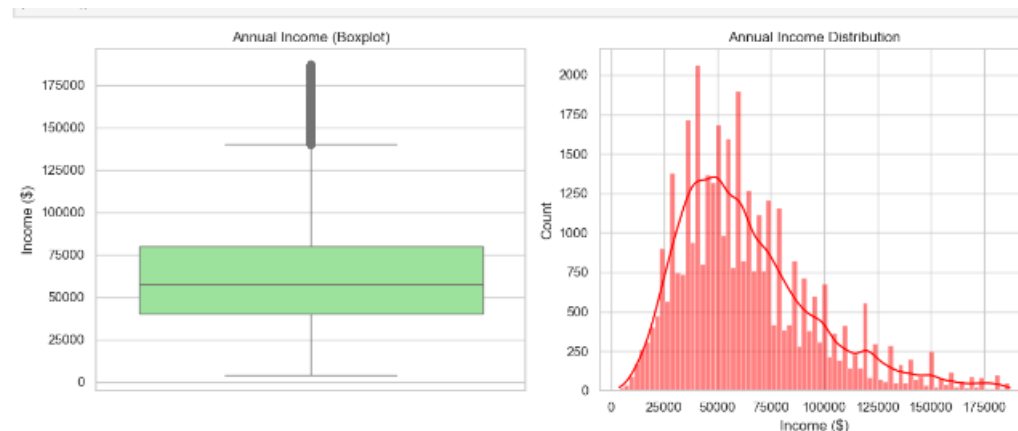
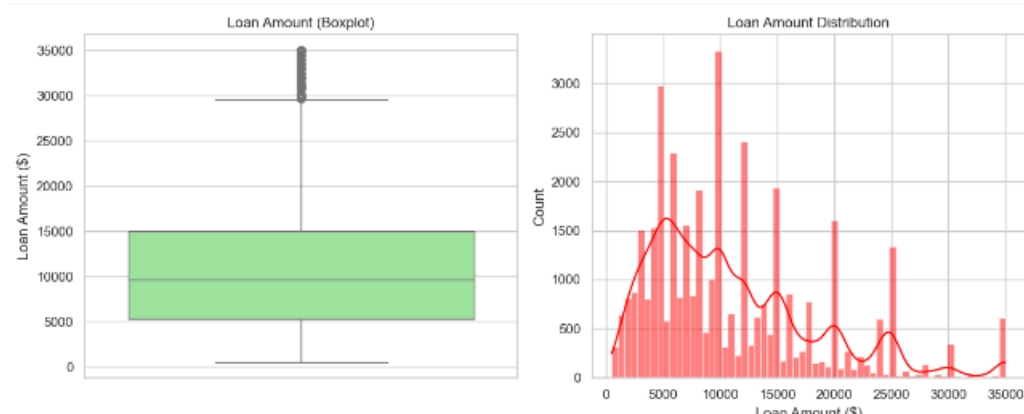
Univariate Analysis: Insights



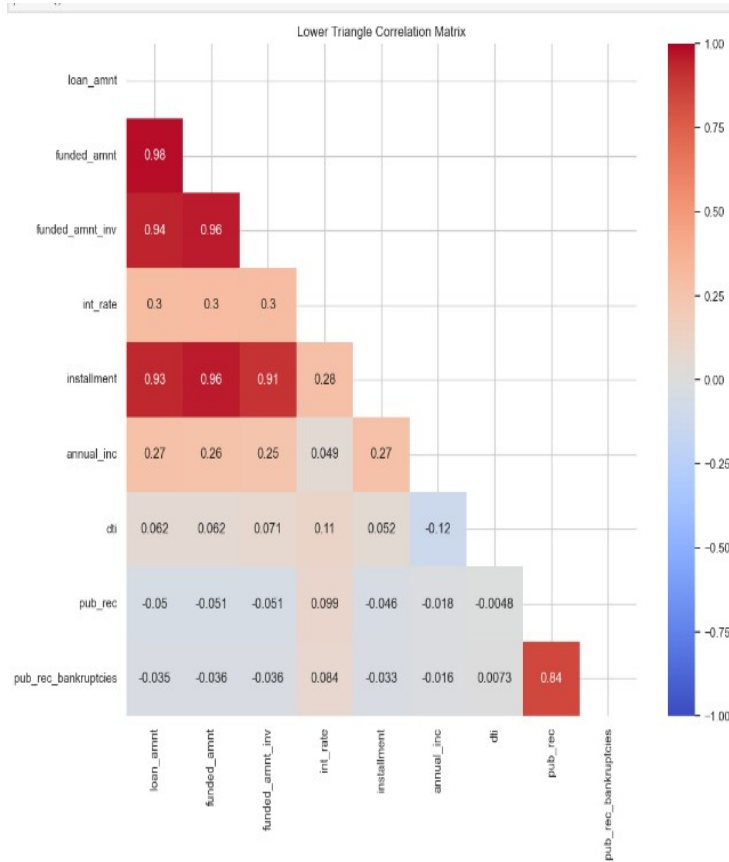
- 'Debt Consolidation' followed by 'Credit card', are the top two reasons for loan applications amongst approved loans.
- California has more than twice as many approved applicants as compared to any other state

Univariate Analysis: Insights

- 50% of loan applicants request a loan amount between 5.3K USD and 15K USD.
- Annual Income was highly skewed as expected; ignoring the outliers, the average salary of applicants is 68.78K USD.
- 50% of the applicants earn between 40K USD and 82K USD annually



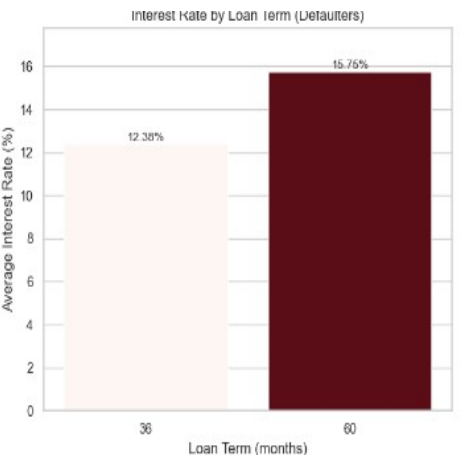
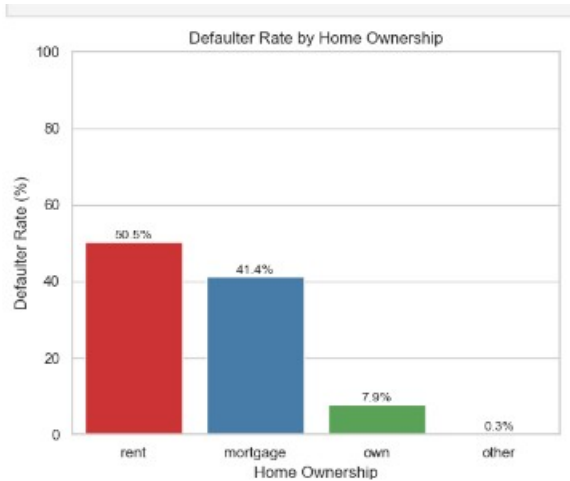
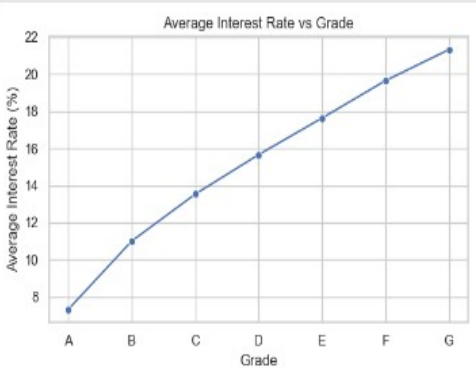
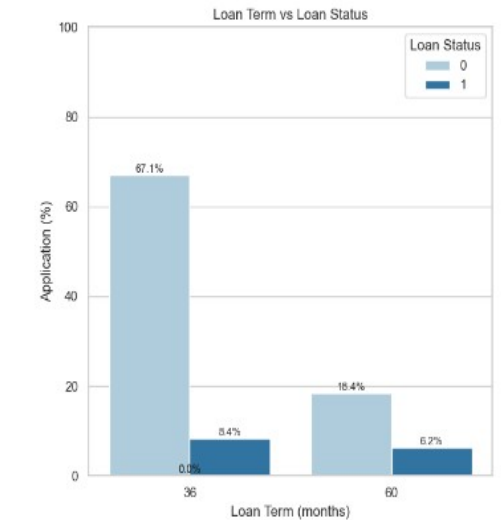
Bivariate Analysis



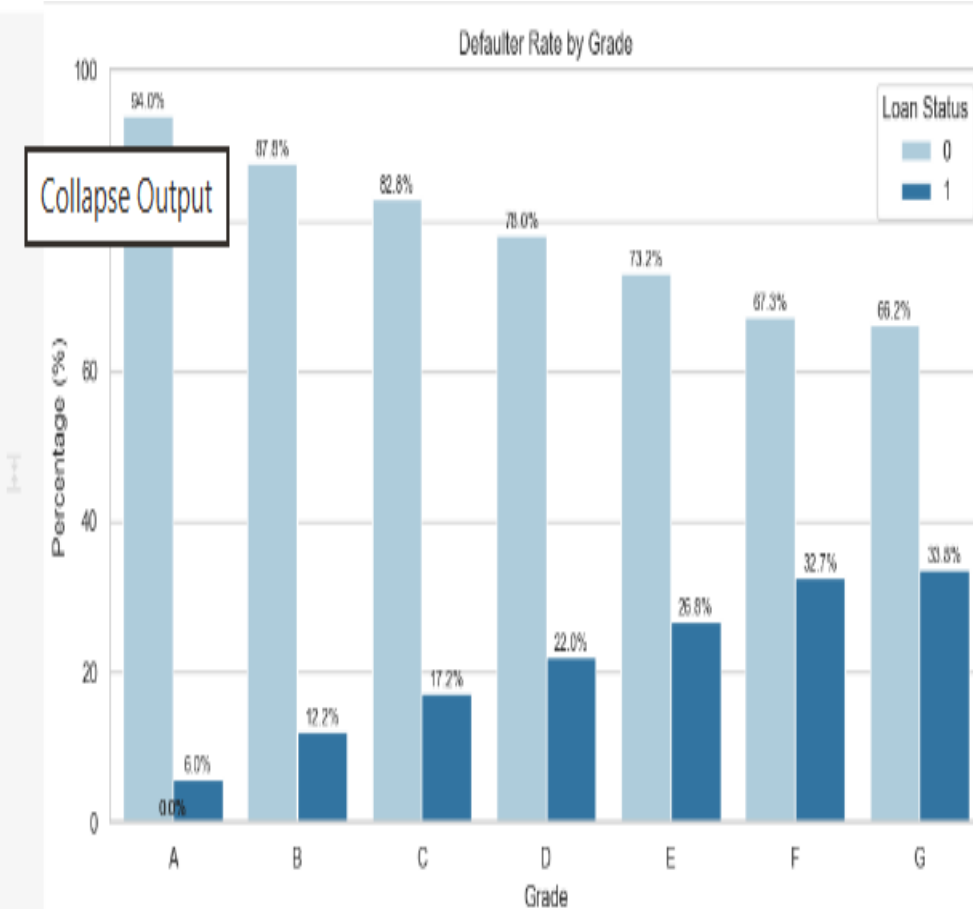
- Pearson Correlation plot between numeric/continuous variables retained
- Numbers are indicative of the strength of correlation between variable

Bivariate Analysis

- Average Interest rates increase as the risk associated with the applicants increase, which can be denoted by the grades assigned to applications. It is an effective process to follow.
- Loan approved for 36 months seems to have a greater % of defaulters as compared to loans approved for 60 months term.
- 91 % of the defaulters already have a mortgage taken out or live on rent.
- Average Interest rate for defaulted applications is very high with 12.38 % for 36 months and 15.75 % for 60 months term

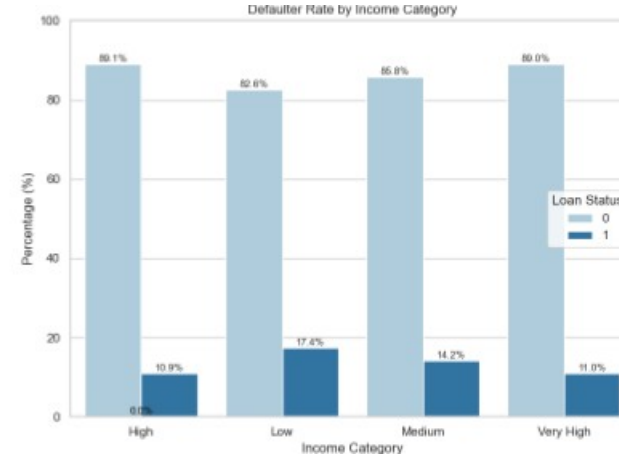
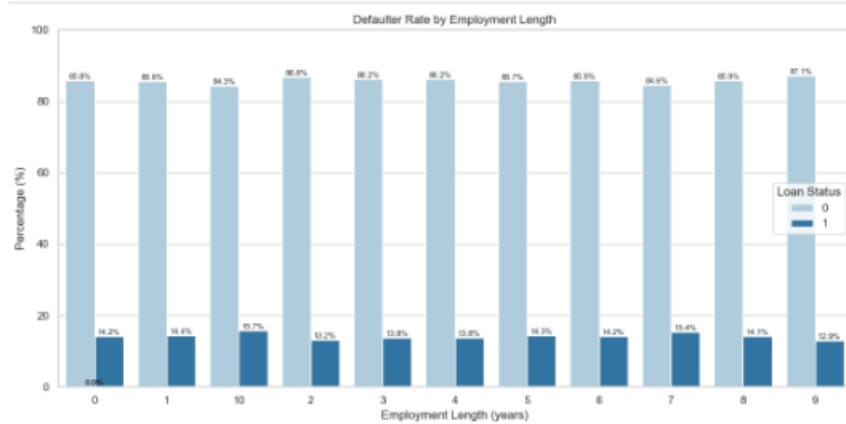


Bivariate Analysis: Grade Vs Loan Status



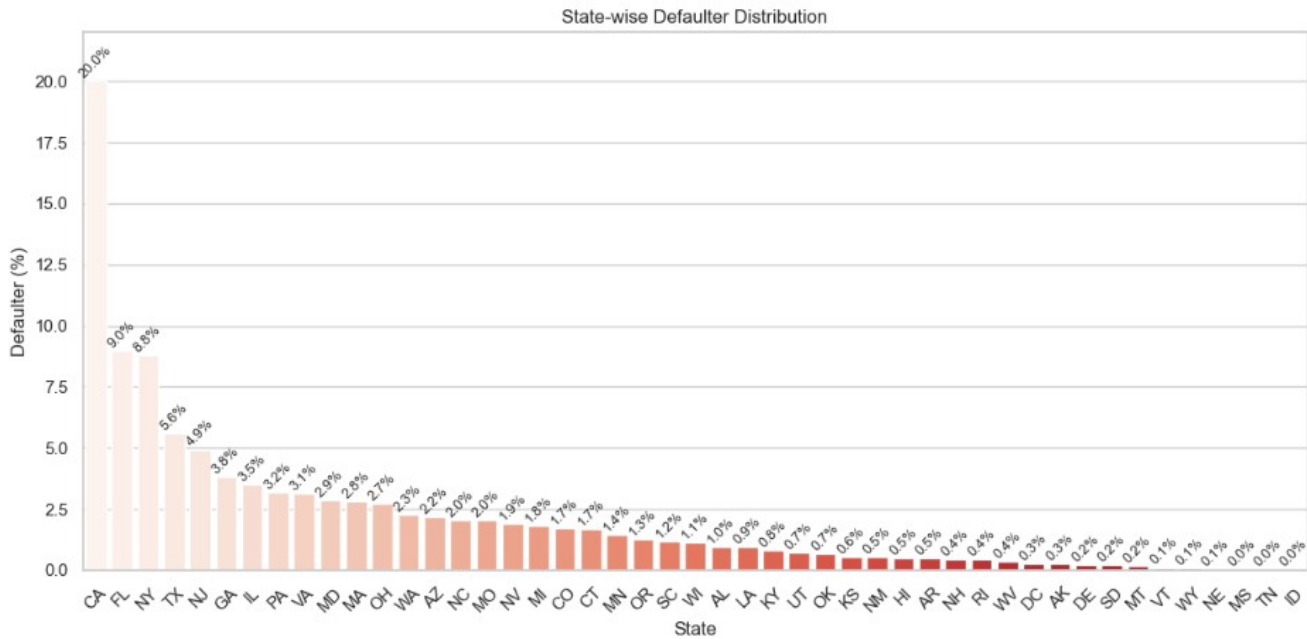
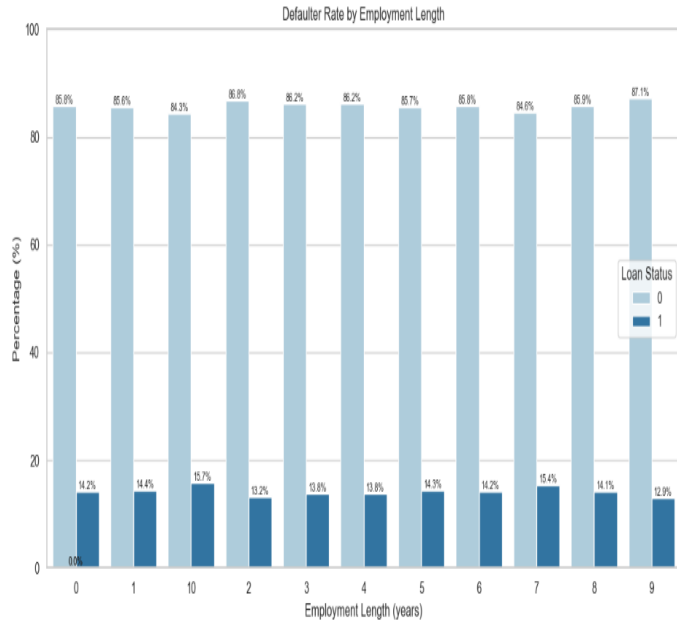
- Percentage of defaulters is significantly high within the grade categories, 'F' and 'G'. LC should be more vigilant while approving loan applications for high risk applicants.
- Largest Proportion of approved loan defaulters belong to Grade 'B'(3.7%) and 'C'(3.5%) category

Bivariate Analysis: Loan Amount & Income Vs Loan Status



- Binning of Requested Loan amount clearly shows that 'defaulter rate' increases as the requested loan amount Increases
- Loan applicants from 'Low'(<=45K USD) and 'Medium'(45K 90K USD) income group have a greater share of defaulted loans

Bivariate Analysis: Loan Status Vs Grade & State



- Maximum percentage of defaulters are found in the 10 years (4%) employment year category. After it, applicants with less years of experience i.e. around 0 to 2 years, following the trend.
- We see An interesting insight shows that almost 41% of defaulters hail from just 4 states namely California, New York , Florida and Texas.

Conclusion:

Driving Factors (or driver variables):

- **Loan Term** : Average Interest rate for defaulted applications is very high with 12.38 % for 36 months and 15.75 % for 60 months term.
- **Grade** : Default Rate is high in high risk loan applicants. It would be important for LC to thoroughly vet high risk loan applicatio
- **Loan Amount** : Defaulter rate increases as the requested loan amount increases.
- **Annual Income** : Applicants from ▪ **Employment Length** ns. 'Low'(<=45K USD) and 'Medium'(45K 90K USD) income group have a greater share of defaulted loans. : Maximum number of defaulters have 10/10+ years of experience and 0 to 2 years of experience. Hence, LC should be take this aspect into consideration while lending loans.
- **Loan Purpose** : The top two reasons for loans are debt consolidation and credit card. Such applications should be carefully assessed